

This is mediated by the inhibition of the p38 MAPK-MAFbx and NF- $\kappa$ B-MuRF-1 pathways. Our data point out the potential benefit of allopurinol and indomethacin administration for bedridden, astronauts, sarcopenic and cachexic patients.

<http://dx.doi.org/10.1016/j.freeradbiomed.2014.10.799>

## Redox Biology Poster Prize Winner

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### Detection and quantification of protein oxidation in sarcopenic models: a mass spectrometry study

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#### Abstract

Oxidised biomolecules in aged tissue could potentially be used as biomarkers for age-related diseases; however, it is still unclear whether they causatively contribute to ageing or are consequences of the ageing process. To assess the potential of using protein oxidation as markers of ageing, mass spectrometry (MS) was employed for the identification and quantification of oxidative modifications in obese (ob/ob) mice. Lean muscle mass and strength is reduced in obesity, representing a sarcopenic model in which the levels of oxidation can be evaluated for different muscular systems including calcium homeostasis, metabolism and contractility. Several oxidised residues were identified by tandem MS (MS/MS) in both muscle homogenate and isolated sarcoplasmic reticulum (SR), an organelle that regulates intracellular calcium levels in muscle. These modifications include oxidation of methionine, cysteine, tyrosine, and tryptophan in several proteins such as sarcoplasmic reticulum calcium ATPase (SERCA), glycogen phosphorylase, and myosin. Once modifications had been identified, multiple reaction monitoring MS (MRM) was used to quantify the percentage modification of oxidised residues within the samples. Preliminary data suggests proteins in ob/ob mice are more oxidised than the controls. For example SERCA, which constitutes 60–70% of the SR, had approximately a 2-fold increase in cysteine trioxidation of Cys561 in the obese model when compared to the control. Other obese muscle proteins have also shown a similar increase in oxidation for various residues. Further analysis with complex protein mixtures will determine the potential diagnostic use of MRM experiments for analysing protein oxidation in small biological samples such as muscle needle biopsies.

<http://dx.doi.org/10.1016/j.freeradbiomed.2014.10.800>

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### Pancreatic ascites haemoglobin up-regulates the HIF/VEGF pathway in the lung in severe acute pancreatitis

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#### Abstract

Extracellular haemoglobin (EHb) is considered a toxic molecule due to its cytotoxicity and peroxidase activity. EHb may release free hemin that

increases vascular permeability, leukocyte recruitment, and adhesion molecule expression. Pancreatitis-associated ascitic fluid is reddish and may contain cell-free hemoglobin. Our aim was to determine the role of EHb in the local and systemic inflammatory response during severe acute pancreatitis in rats. To this end, taurocholate-induced necrotizing pancreatitis in rats was used. EHb levels were quantified in ascites and plasma and the hemolytic action of ascitic fluid was tested. Furthermore, we assessed if peritoneal lavage prevented the increase in EHb levels in plasma during pancreatitis. Finally, hemoglobin was purified from rat erythrocytes and administered i.p. to assess the local and systemic effects of ascitic-associated EHb during acute pancreatitis. EHb levels markedly increased in ascitic fluid and plasma during necrotizing pancreatitis. Peroxidase activity was very high in ascites. The peritoneal lavage abrogated the increase in cell-free hemoglobin in plasma. The administration of EHb enhanced ascites, dramatically increased abdominal fat necrosis, up-regulated *tumor necrosis factor- $\alpha$* , *interleukin 1 $\beta$*  and *interleukin 6* gene expression and decreased expression of *interleukin 10* in abdominal adipose tissue during pancreatitis. EHb enhanced the gene expression and protein levels of vascular endothelial growth factor (VEGF) and other hypoxia inducible factor-related genes [*inducible nitric oxide synthase (inos)*, *endothelial nitric oxide synthase (enos)* and *hexokinase 2*] in the lung. EHb also increased myeloperoxidase activity in the lung. In conclusion, extracellular hemoglobin contributes to the inflammatory response in severe acute pancreatitis through abdominal fat necrosis and inflammation and increasing VEGF and leukocyte infiltration in the lung.

<http://dx.doi.org/10.1016/j.freeradbiomed.2014.10.801>

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### The Relations Between Immunity, Oxidative Stress and Inflammation Markers, in Childhood Obesity

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#### Abstract

Oxidative stress, inflammation and insulin resistance are the principal culprits in childhood obesity. Immune modifications are also important in the development of the obesity complications. The aim of this study is to find the relations for some immunity parameters with markers for oxidative stress and inflammation.

Sixty obese children (10–16 years old) and thirty age and sex matched lean children were involved. The activities for erythrocyte superoxide dismutase (SOD), for erythrocyte glutathione peroxidase (GPx) and serum thioredoxin level were measured by ELISA, as oxidative stress markers. Circulating immune complexes (CIC), complement fractions C3, C4 and the self-antibodies, antismooth muscle antibodies (ASMA), antiliver-kidney microsome antibodies (LKM1) were measured by ELISA methods. Ceruloplasmin, haptoglobin and C reactive protein (CRP) were measured as inflammatory markers by immunoturbidimetric methods.

ceruloplasmin ( $p < 0.001$ ), haptoglobin ( $p < 0.001$ ), CRP ( $p < 0.05$ ) and activity for SOD ( $p < 0.001$ ) were measured, while thioredoxin concentration ( $p < 0.04$ ) was reduced. The antibodies LKM1 and ASMA and GPx activity were not modified between groups. Positive correlations (for  $p < 0.05$ ) were calculated between SOD activity and LKM1 ( $r = 0.37$ ), GPx activity and ASMA ( $r = 0.27$ ), haptoglobin and C3 ( $r = 0.33$ ), ceruloplasmin and CIC ( $r = 0.41$ ), CRP and C3 ( $p < 0.27$ ) and negative correlations were calculated for C4 both with GPx activity ( $r = -0.28$ ) and with thioredoxin level ( $r = -0.27$ ).

In the obese children versus the lean ones, higher levels for C3 ( $p < 0.001$ ), C4 ( $p < 0.001$ ), CIC ( $p < 0.05$ ). In conclusion, this study demonstrates that